Week 94: Calibration lab

Adult guide built from the problem files · Unpiloted. Printing, cutting, handling and the games are unrehearsed.

This week's adult guide is week-94-facilitator.pdf, whose source is editable/guide/facilitator.tex. These problem files hold each problem's one-sentence idea, not its answers.

Problems and routes

Problem	K–1	2–3	4–5	route
How long is each strip?	P1			first
Every place with one end at 7	P2			first
Two strips covered by a third	P3			first
Play the extra-number machine	P4	P1		first
Every pair of hidden lengths	P5	P2		first
What you can know without the extra		P3		first
A known strip finds the hidden strips	P6	P4		first
The covered-lengths game		P5		first
Which boxes could be right?		P6		return
Make a one-wrong-report puzzle		P7		return
Play the copy-and-add machine			P1	first
A 2-unit strip reports 6			P2	first
Two reports fix a rule			P3	first
Repair three reports			P4	first
Tests without a promised rule			P5	first
Smallest and largest stretch			P6	return
Choose the best references			P7	return

First route. K–1 P1–P6; Grades 2–3 P1–P5; Grades 4–5 P1–P5. **Return visits.** K–1 none; Grades 2–3 P6–P7; Grades 4–5 P6–P7.

Key, one entry per problem

How long is each strip?

K–1 P1 · first route

Idea. On a ruler with no zero, a length is the number of spaces between the ends, not the label at the right end.

Every place with one end at 7

K–1 P2 · first route

Idea. A strip with one end at 7 can run either way, unless it runs off the ruler: D fits only from 7 to 13.

Two strips covered by a third

K–1 P3 · first route

Idea. A and C end to end match D exactly: $2 + 4 = 6$; no other pair of different strips matches a strip.

Play the extra-number machine

K–1 P4 · Grades 2–3 P1 · first route · wording differs by band

Idea. One known strip always reveals the extra: report minus length. In K–1, the tower that comes out stands taller than its input by the extra.

Every pair of hidden lengths

K–1 P5 · Grades 2–3 P2 · first route · wording differs by band

Idea. Reports 7 and 10 fit six pairs, one for each extra: (6, 9), (5, 8), (4, 7), (3, 6), (2, 5), (1, 4). In K–1, towers that come out as 4 and 6 with extras 1–3 went in as (3, 5), (2, 4) or (1, 3); Q is always 2 taller.

What you can know without the extra

Grades 2–3 P3 · first route

Idea. Q is always exactly 3 longer than P: the shared extra cancels in the difference.

A known strip finds the hidden strips

K–1 P6 · Grades 2–3 P4 · first route · wording differs by band

Idea. A known strip fixes the extra (4, 2, 6), and then the hidden lengths. In K–1, the known towers give extras 1, 3 and 2, so P and Q went in as (3, 5), (1, 3) and (2, 4).

The covered-lengths game

Grades 2–3 P5 · first route

Idea. Any one known strip reveals the extra and so both hidden lengths.

Which boxes could be right?

Grades 2–3 P6 · return route

Idea. Only box II has one extra throughout; I and III each have one repair.

Make a one-wrong-report puzzle

Grades 2–3 P7 · return route

Idea. Three pairs with one wrong report always have exactly one repair.

Play the copy-and-add machine

Grades 4–5 P1 · first route

Idea. Two different known strips always identify the machine; a single request can, with luck or a length of 7 or more.

A 2-unit strip reports 6

Grades 4–5 P2 · first route

Idea. 2 units to 6 leaves (1, 4), (2, 2) and (3, 0); any second length other than 2 separates them.

Two reports fix a rule

Grades 4–5 P3 · first route

Idea. Two exact reports fix a stretch-and-shift rule: A is $y = 2x + 4$, B is $y = x + 6$.

Repair three reports

Grades 4–5 P4 · first route

Idea. The three reports are not on one line; there are exactly three single-report repairs.

Tests without a promised rule

Grades 4–5 P5 · first route

Idea. With no promised form, no finite set of tests forces the report at an untested length.

Smallest and largest stretch

Grades 4–5 P6 · return route

Idea. Pair A allows stretches 1 to 3; pair B, farther apart, 7/4 to 9/4.

Choose the best references

Grades 4–5 P7 · return route

Idea. With errors up to 1, references D apart leave stretches in a range at most $4/D$ wide: 1 and 10 are best, and one repeated reference never pins the stretch.